Light

GCSE Computer Science – Searching and Sorting Algorithms (Set 2)

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GCSE • AQA 8525 • Grades 3—9 | Topic test | 30 min | 55 marks | Answered: 0 / 30

00:00 Start Pause Reset ← Back to Algorithms Searching Sorting

Multiple-choice topic test for **GCSE Computer Science (AQA 8525)**. You may skip questions. When you submit, you'll see your score, a grade estimate, and worked explanations. **Tip:** use the stopwatch above to time yourself.

Q1 (1 mark)

In the worst case scenario for a linear search, where would the target item be located?

- ○ **A.** At the very beginning of the list
- ○ **B.** In the middle of the list
- ○ **C.** At a random position each time
- ○ **D.** At the very end of the list, or not present at all

Q2 (1 mark)

A student wants to use binary search on their contact list. What must be true about the contact list before binary search can work correctly?

- ○ **A.** The contacts must be arranged in alphabetical order
- ○ **B.** The contacts must all have phone numbers
- ○ **C.** The list must contain exactly 100 contacts
- ○ **D.** The contacts must be added within the last year

Q3 (1 mark)

In bubble sort, which items are compared during the sorting process?

- ○ **A.** The first item and the last item
- ○ **B.** Every item with every other item simultaneously
- ○ **C.** Items that are next to each other (adjacent pairs)
- ○ **D.** Only the largest and smallest items

Q4 (1 mark)

In merge sort, why is a list containing just one item considered to be already sorted?

- ○ **A.** Because single items are always the largest
- ○ **B.** Because the computer processes single items faster
- ○ **C.** Because single items are deleted during sorting
- ○ **D.** Because there is nothing to compare it with, so no items can be out of order

Q5 (1 mark)

What is one key advantage of linear search over binary search?

- ○ **A.** Linear search is always faster
- ○ **B.** Linear search works on data that is not sorted
- ○ **C.** Linear search uses less memory than binary search
- ○ **D.** Linear search can only find one item

Q6 (1 mark)

After completing one full pass of bubble sort in ascending order, which item is guaranteed to be in its correct final position?

- ○ **A.** The smallest item in the list
- ○ **B.** The middle item in the list
- ○ **C.** All items are in their correct positions
- ○ **D.** The largest item in the unsorted portion

Q7 (1 mark)

What happens during the 'divide' phase of merge sort?

- ○ **A.** The list is repeatedly split in half until each part contains one item
- ○ **B.** Items are compared and swapped
- ○ **C.** Sorted sublists are combined together

- ○ **D.** The largest items are moved to the end

Q8 (1 mark)

When performing a binary search, which item is always examined first?

- ○ **A.** The first item in the list
- ○ **B.** The last item in the list
- ○ **C.** The middle item in the list
- ○ **D.** A randomly selected item

Q9 (1 mark)

When comparing algorithm efficiency at GCSE level, what aspect of efficiency is considered?

- ○ **A.** How much electricity the algorithm uses
- ○ **B.** How colourful the output appears
- ○ **C.** How many programmers are needed to write it
- ○ **D.** How much time the algorithm takes to complete

Q10 (1 mark)

In the best case scenario for linear search, how many comparisons are needed to find the target item?

- ○ **A.** One comparison

- B. Zero comparisons
- C. Half the list needs to be checked
- D. The entire list needs to be checked

Q11 (1 mark)

During the merge phase of merge sort, what operation occurs?

- A. The list is split into smaller parts
- B. Sorted sublists are combined into larger sorted sublists by comparing items
- C. Items are randomly shuffled
- D. Only the first and last items are swapped

Q12 (1 mark)

How does binary search achieve faster searching than linear search on sorted data?

- A. By eliminating half of the remaining items with each comparison
- B. By checking every item twice
- C. By sorting the data first
- D. By using more memory

Q13 (2 marks)

A linear search is performed on the list `[15, 8, 22, 7, 13]` to find the value 22. How many comparisons are needed?

- ○ **A.** 1 comparison
- ○ **B.** 2 comparisons
- ○ **C.** 3 comparisons
- ○ **D.** 5 comparisons

Q14 (2 marks)

A binary search is performed on the sorted list `[4, 9, 14, 20, 27, 33, 41]` to find the value 33. After comparing with the middle item, which direction does the search proceed?

- ○ **A.** The search ends because 33 is the middle item
- ○ **B.** Left, because 33 is less than the middle item
- ○ **C.** The search starts over from the beginning
- ○ **D.** Right, because 33 is greater than the middle item

Q15 (2 marks)

The list `[9, 3, 7, 2]` undergoes one complete pass of bubble sort in ascending order. What is the resulting list?

- ○ **A.** [2, 3, 7, 9]
- ○ **B.** [3, 7, 2, 9]
- ○ **C.** [3, 9, 7, 2]

- **D.** [9, 3, 2, 7]

Q16 (2 marks)

When merge sort divides the list `[5, 2, 8, 1, 9, 4]`, how many levels of splitting are needed to reach individual items?

- **A.** 1 level
- **B.** 6 levels
- **C.** 3 levels
- **D.** 2 levels

Q17 (2 marks)

A database contains 10,000 sorted customer records. Which search algorithm would find a specific customer record using fewer comparisons?

- **A.** Linear search, because it checks every record systematically
- **B.** Both would use the same number of comparisons
- **C.** Neither can search 10,000 records
- **D.** Binary search, because it can eliminate half the records with each comparison

Q18 (2 marks)

During bubble sort, a complete pass is made through the list with zero swaps occurring. What does this indicate?

- ○ **A.** The list is now fully sorted
- ○ **B.** The algorithm has failed
- ○ **C.** The list needs to be reversed
- ○ **D.** More passes are definitely needed

Q19 (2 marks)

When merge sort merges the sorted sublists `[3, 9]` and `[4, 7]`, what is the first item placed in the output?

- ○ **A.** 9 (the largest item)
- ○ **B.** 4 (the first item of the second sublist)
- ○ **C.** 3 (the smallest item from the first sublist)
- ○ **D.** 7 (a random selection)

Q20 (2 marks)

In binary search, if the current search range spans from index 4 to index 12, which index is examined next?

- ○ **A.** Index 4
- ○ **B.** Index 8
- ○ **C.** Index 12
- ○ **D.** Index 6

Q21 (2 marks)

A list contains 6 items. What is the maximum number of passes bubble sort might need to fully sort the list?

- ○ **A.** 3 passes
- ○ **B.** 5 passes
- ○ **C.** 6 passes
- ○ **D.** 36 passes

Q22 (2 marks)

A company's inventory list has items added and removed frequently throughout the day. Why might linear search be more practical than binary search for this list?

- ○ **A.** Linear search is always faster regardless of list organisation
- ○ **B.** Binary search cannot search for products
- ○ **C.** Linear search only works on frequently changing data
- ○ **D.** Keeping the list sorted for binary search would require constant re-sorting after every change

Q23 (2 marks)

What is one disadvantage of merge sort compared to bubble sort?

- ○ **A.** Merge sort requires additional memory to store sublists during the merging process
- ○ **B.** Merge sort is always slower than bubble sort

- ○ **C.** Merge sort cannot sort numbers
- ○ **D.** Merge sort only works on lists with exactly 8 items

Q24 (3 marks)

A binary search is performed on the sorted list `[5, 11, 18, 24, 31, 39, 45, 52, 60]` to find the value 39. Trace the complete search showing all comparisons made.

- ○ **A.** Compare 31 → go right → Compare 52 → go left → Compare 39 (found) - 3 comparisons
- ○ **B.** Compare 39 (found) - 1 comparison
- ○ **C.** Compare 5 → Compare 11 → Compare 18 → Compare 24 → Compare 31 → Compare 39 (found) - 6 comparisons
- ○ **D.** Compare 31 → go right → Compare 45 → go left → Compare 39 (found) - 3 comparisons

Q25 (3 marks)

Trace bubble sort completely on the list `[7, 3, 5, 2]` sorting into ascending order. How many swaps are performed in total across all passes?

- ○ **A.** 4 swaps total
- ○ **B.** 3 swaps total
- ○ **C.** 6 swaps total
- ○ **D.** 5 swaps total

Q26 (3 marks)

Trace merge sort on the list `[8, 4, 6, 2]` completely. What are the intermediate sublists immediately before the final merge?

- ○ **A.** [8, 4] and [6, 2]
- ○ **B.** [4, 8] and [2, 6]
- ○ **C.** [2, 4] and [6, 8]
- ○ **D.** [4, 6] and [2, 8]

Q27 (3 marks)

A list of 64 items is searched. Compare the maximum number of comparisons needed for linear search versus binary search in the worst case.

- ○ **A.** Linear: 64, Binary: 64
- ○ **B.** Linear: 32, Binary: 6
- ○ **C.** Linear: 6, Binary: 64
- ○ **D.** Linear: 64, Binary: 6

Q28 (3 marks)

A school needs to sort a list of 200 student names for a graduation programme. The names are currently in random order. Analyse which sorting algorithm would likely be more time-efficient.

- ○ **A.** Bubble sort, because it is simpler to understand
- ○ **B.** Merge sort, because its divide-and-conquer approach handles

200 items more efficiently than bubble sort's repeated passes

- ○ **C.** They would take exactly the same time
- ○ **D.** Neither algorithm can sort 200 items

Q29 (3 marks)

A programmer attempts to use binary search on the unsorted list `[12, 3, 18, 7, 9]` to find the value 3. Explain why the algorithm fails to find 3 even though it is present.

- ○ **A.** Binary search starts at position 0 and checks 12 first, missing 3
- ○ **B.** Binary search automatically sorts the list first, then finds 3
- ○ **C.** The algorithm finds the middle (18), sees 3 < 18, searches left half [12, 3], then checks middle (12), sees 3 < 12, searches left but nothing is there - incorrectly reporting 3 is not found
- ○ **D.** 3 is too small for binary search to detect

Q30 (3 marks)

A smartphone app maintains a contacts list that users search frequently. The contacts are stored in alphabetical order and rarely change. Evaluate the most appropriate search algorithm and justify the choice.

- ○ **A.** Linear search, because phones have limited processing power
- ○ **B.** No search algorithm is needed since contacts are alphabetical
- ○ **C.** Bubble sort should be used to search the contacts

○ **D.** Binary search, because the data is already sorted, searches are frequent, and changes are rare - maximising the benefit of binary search's efficiency

Total: **55** marks • Time: **30 min**

↑ Back to top

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